Graphing Rational Functions

ANSWER KEY

Reading graphs

1 The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.

Vertical asymptote $x = 3$; horizontal asymptote $y = 1$; intercepts $(-2, 0)$ and $\left(0, -\frac{2}{3}\right)$.

2 The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.

a Describe the transformation. Shift right 2 units and up 3 units.

b State the asymptotes of g. Vertical $x = 2$; horizontal $y = 3$.

c State the domain and range. Domain $x \neq 2$; range $y \neq 3$.

Sign analysis

3 Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.

Testing each interval: $(-\infty, -2)$ negative; $(-2, 1)$ positive; $(1, 4)$ negative; $(4, \infty)$ positive. So $f(x) > 0$ on $(-2, 1) \cup (4, \infty)$.

4 Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.

Critical values $x = -3$ (zero) and $x = 2$ (undefined). The quotient is negative between them: $[-3, 2)$ — including -3, excluding 2.

5 Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.

Vertical asymptotes $x = 1$ and $x = -1$; horizontal asymptote $y = 0$. Since $h(-x) = -h(x)$, the function is odd — symmetric about the origin.

Sketching a complete graph

6 For $f(x) = \frac{x^2-1}{x^2-4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.

x-intercepts: $x = \pm 1$. y-intercept: $f(0) = \frac{-1}{-4} = \frac{1}{4}$. Vertical asymptotes $x = \pm 2$; horizontal asymptote $y = 1$ (equal degrees). Near $x = 2$: $f \to +\infty$ as $x \to 2^+$ and $f \to -\infty$ as $x \to 2^-$. Near $x = -2$: by symmetry (the function is even), $f \to -\infty$ as $x \to -2^+$ and $f \to +\infty$ as $x \to -2^-$.

7 Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its x- and y-intercepts.

y-intercept: $g(0) = \frac{3}{-1} - 2 = -5$. x-intercept: $0 = \frac{3}{x-1} - 2$ gives $\frac{3}{x-1} = 2$, so $x - 1 = \frac{3}{2}$, $x = \frac{5}{2}$.

More rational inequalities

8 Solve $\frac{(x-2)(x+1)}{x-4} \geq 0$.

Critical values $x = -1, 2, 4$. Sign chart: negative on $(-\infty, -1)$, positive on $(-1, 2)$, negative on $(2, 4)$, positive on $(4, \infty)$. Solution (including zeros, excluding the undefined point): $[-1, 2] \cup (4, \infty)$.

9 Solve $\frac{2}{x+3} > 1$.

Rewrite: $\frac{2}{x+3} - 1 > 0 \Rightarrow \frac{2-(x+3)}{x+3} > 0 \Rightarrow \frac{-x-1}{x+3} > 0 \Rightarrow \frac{x+1}{x+3} < 0$. Critical values $-3, -1$: solution $(-3, -1)$.